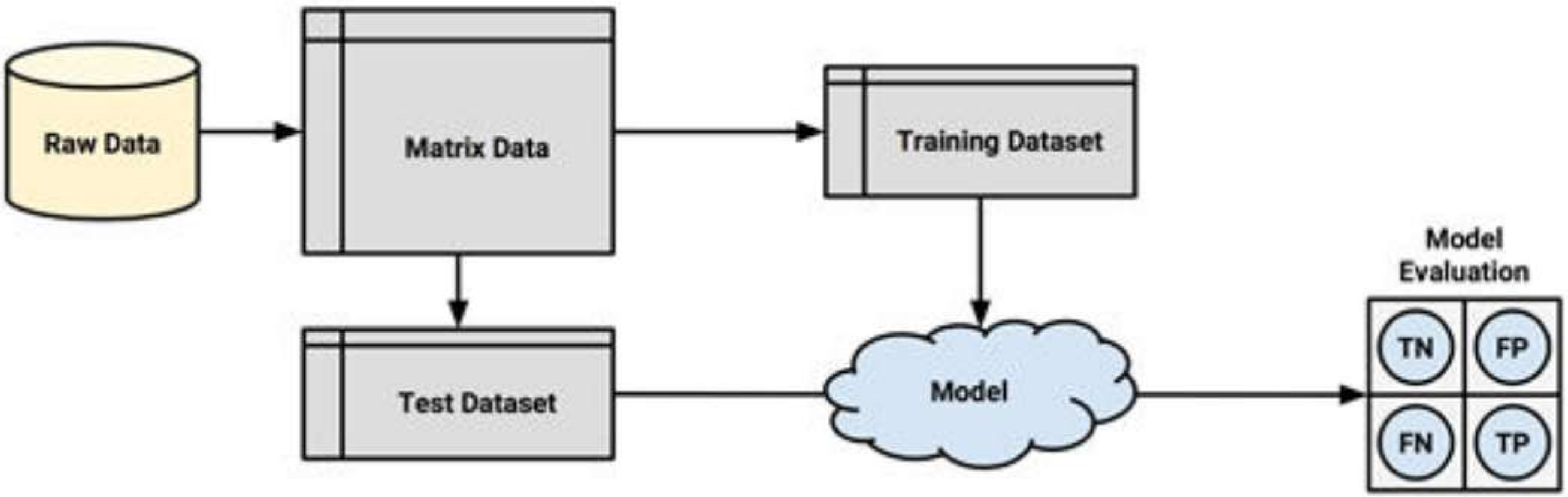
Cross validation

Module-5

The holdout method

The procedure of partitioning data into training and test datasets is known as the holdout method.



The **hold-out method** divides the available dataset into **two or three disjoint subsets**:

1. **Training set** – used to train the model
2. **Validation set (optional)** – used to tune model parameters
3. **Test set** – used to evaluate the model's final performance

A common division is:

$$\text{Training} : \text{Testing} = 70 : 30 \quad \text{or} \quad 80 : 20$$

If a validation set is also used:

$$\text{Training} : \text{Validation} : \text{Testing} = 60 : 20 : 20$$

- One problem with hold out sampling is that each partition may have larger or smaller proportion of some classes
- Small proportion class may be omitted from the training and the model will not be able to learn from the class .
- A technique called repeated holdout is sometimes used to mitigate the problems of randomly composed training datasets.
- The repeated holdout method is a special case of the holdout method that uses the average result from several random holdout samples to evaluate a model's performance. The repeated holdout is the basis of a technique known as **k fold cross-validation (or k-fold CV)**, which has become the industry standard for estimating model performance.

Cross-validation

Cross-validation is a technique in which we train our model using the subset of the data-set and then evaluate using the complementary subset of the data-set.

The three steps involved in cross-validation are as follows :

1. Reserve some portion of sample data-set.
2. Using the rest data-set train the model.
3. Test the model using the reserve portion of the data-set.

• Methods of Cross Validation

Validation

In this method, we perform training on the 50% of the given data-set and rest 50% is used for the testing purpose.

The major drawback of this method is that we perform training on the 50% of the dataset, it may possible that the remaining 50% of the data contains some important information which we are leaving while training our model i.e higher bias.

- **LOOCV (Leave One Out Cross Validation)**

In this method, we perform training on the whole data-set but leaves only one data-point of the available data-set and then iterates for each data-point.

An advantage of using this method is that we make use of all data points and hence it is low bias.
The major drawback of this method is that it leads to higher variation in the testing model as we are testing against one data point.

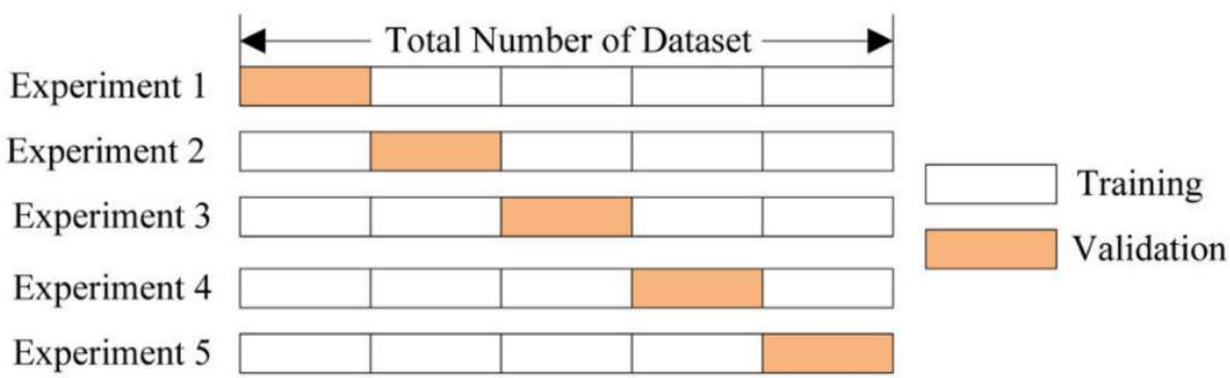
If the data point is an outlier it can lead to higher variation.

Another drawback is it takes a lot of execution time as it iterates over ‘the number of data points’ times.

- **K-Fold Cross Validation**
- In this method, we split the data-set into k number of subsets(known as folds) then we perform training on the all the subsets but leave one(k-1) subset for the evaluation of the trained model.
- In this method, we iterate k times with a different subset reserved for testing purpose each time.

- The diagram shows an example of the training subsets and evaluation subsets generated in k-fold cross-validation. Here, we have total 25 instances. In first iteration we use the first 20 percent of data for evaluation, and the remaining 80 percent for training([1-5] testing and [5-25] training) while in the second iteration we use the second subset of 20 percent for evaluation, and the remaining three subsets of the data for ... d so on.

5 fold CV



- Advantages of train/test split:

1. This runs K times faster than Leave One Out cross-validation because K-fold cross-validation repeats the train/test split K-times.
2. Simpler to examine the detailed results of the testing process.

- Advantages of cross-validation:

1. More accurate estimate of out-of-sample accuracy.
2. More “efficient” use of data as every observation is used for both training and testing.